Jisoo Kim

email: jk838@cam.ac.uk website: https://jisooneuro.github.io/
Google scholar PubMed ORCID

EDUCATION

PhD in Neuroscience, *University of Cambridge, UK.* Oct. 2021 - Present
Thesis submitted (November 2025)

Masters in Brain and Cognitive Engineering, *Korea University, South Korea.* Sep. 2018 – Aug. 2020

Bachelor in Bio-system Medical Science, *Korea University, South Korea* Mar. 2014 – Aug. 2018

Exchange student, *University of Copenhagen, Denmark* Dec. 2016 – July 2017

PUBLICATIONS

Jisoo Kim, Riccardo Beltramo, Jasper Poort
"Contributions of superior colliculus and primary visual cortex to visual spatial detection in freely moving mice."
Current Biology (2026) https://doi.org/10.1016/j.cub.2026.01.077

Seoung-Jun Jo*, Hio-Been Han*, Dayoung Jung, **Jisoo Kim**, Jee Hyun Choi
"Mouse Escape Behaviors and mPFC-BLA Activity Dataset: Understanding Flexible Defensive Strategies Under Threat." ***Scientific Data.*** (2024) https://doi.org/10.1038/s41597-024-03688-0

Ana González-Rueda, Kristopher Jensen, Mohammadreza Noormandipour, Daniel de Malmazet, Jonathan Wilson, Ernesto Ciabatti, **Jisoo Kim**, Elena Wiliams, Jasper Poort, Guillaume Hennequin and Marco Tripodi
"Kinetic features dictate sensorimotor alignment in the superior colliculus."
Nature (2024) https://doi.org/10.1038/s41586-024-07619-2

Hio-Been Han, Hee-Sup Shin, Yong Jeong, **Jisoo Kim**, Jee Hyun Choi
"Dynamic switching of neural oscillations in the prefrontal–amygdala circuit for naturalistic freeze-or-flight."
PNAS (2023) https://doi.org/10.1073/pnas.2308762120

Jisoo Kim*, Chaewoo Kim, Hiobeen-Han, Jocheol-Jun, Wooseob Yeom, Sung Q Lee*, Jee Hyun Choi*,
"A bird's eye view of brain activity in socially interacting mice through mobile edge computing (MEC)"
Science Advances (2020) https://doi.org/10.1126/sciadv.abb9841

* Equally contributed as first authors

POSTER & ORAL PRESENTATIONS

Poster presentations

Jisoo Kim, Riccardo Beltramo, Jasper Poort ***FENS****, Barcelona,* 2026
"Contributions of superior colliculus and visual cortex to visual spatial detection in freely moving mice."
Federation of European Neuroscience Societies (FENS)

Gabriela Pinheiro, **Jisoo Kim**, Jasper Poort ***BBRSC****, Cambridge, 2026*
"The role of V1 and SC for visually guided decision making"
Biotechnology and Biological Sciences Research Council (BBRSC) Symposium

Jisoo Kim, Riccardo Beltramo, Jasper Poort ***COSYNE****, Lisbon,* 2026
"Contributions of superior colliculus and visual cortex to visual detection in freely moving mice."
Computational and Systems Neuroscience (COSYNE)

Jisoo Kim, Chaewoo Kim, Sung Q Lee, Jeehyun Kwag, Jee Hyun Choi ***Keystone***, Daejon, 2020
"Secured inside in a group: revealing a neurological feature of defensive aggregation under threat"
Keystone (*cancelled due to COVID19)

Jisoo Kim, Chaewoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***SfN****, Chicago, 2019*
"Gamma oscillations in basolateral amygdala during escape behaviors: Observations in a group"

Society for Neuroscience (SfN)

Chaewoo Kim, **Jisoo Kim**, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***SfN**, Chicago, 2019*
"Portraits of dominance hierarchy within a group of mice in their trajectories"
Society for Neuroscience (SfN)

Jisoo Kim, Chaewoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***IBRO**, Daegu, 2019*
"Gamma oscillations in basolateral amygdala: individual mouse and collective mice"
International Brain Research Organization (IBRO)

Chaewoo Kim, **Jisoo Kim**, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***IBRO**, Daegu, 2019*
"Measuring dominance hierarchy within a mouse group in an ecologically friendly environment"
International Brain Research Organization (IBRO)

Jisoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***BSI**, Hongcheon, 2018*
"CBRAIN (Collective brain rhythmical activity interaction neuroplatform)"
Korea Institute of Science and Technology (KIST) Brain Science Institute (BSI)

Jisoo Kim, Jeeyun Kim, Shihwa Park, Eunhang Cha, Jeehee Han ***KSCBP**, Jeju, 2016*
"Cross-modal conflict affects subliminal instrumental conditioning"
Korean Society for Cognitive and Biological Psychology (KSCBP)

Oral presentation

• Won the 2nd prize for the best oral presentation ***BSI**, Hongcheon, 2018*

Jisoo Kim, Chaewoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi
"Collective brain rhythmical activity interaction neuroplatform"
Korea Institute of Science and Technology(KIST) Brain Science Institute (BSI)

FUNDING & AWARD

University of Cambridge, Wolfson college-PDN Studentship, fully funded for 4 years £25,000 per annum

School of Biological Science Doctoral Training Program Award, fully funded for 4 years £23,500 per annum

INVITED TALKS

May 2026 *University College London, Institute for Behavioural Neuroscience (IBN), UK*
"Neural Activity During Naturalistic Behavior in Freely Moving Mice"

March 2025 *Korea University, New-Learn (Online)*
"Guidance on Research Publishing and Pursuing International PhD Programs"

March 2024 *University of Cambridge, PDN Chalks and Talks, UK*
"Contributions of the Primary Visual Cortex and Superior Colliculus During a Visual Detection Task"

June 2021 *Korean Association for Consciousness Sciences (KACS) (Online)*
"Insights into Academic Publishing and PhD Application Strategies"

TEACHING EXPERIENCE

University of Cambridge, *demonstrator*

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye Nov. 2024 – Dec. 2024

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye Nov. 2025 – Dec. 2025

LEADERSHIP & VOLUNTEERING

Training and Co-supervision, Trained and co-supervised PhD students in collaborative projects in Poortlab

Committee member *of the 2023 PDN Symposium, University of Cambridge* Dec. 2021 – Apr. 2022

- Collaborated with researchers from diverse backgrounds to organize social programs

Vice president *of Neuroscience society, New-Learn, Korea University* Mar. 2015 – Dec. 2016

- Project team leader of the human psychology project and presented findings at a conference
- Participated in a brain computer interface (BCI) project, controlling a drone through brain wave analysis

Project leader of *Neuroscience society, New-Learn, Korea University* Sep. 2014 – Mar. 2015

- Proposed a research project on subliminal instrumental conditioning and led the research team
- Presented a poster at the Korean Conference on Brain and Psychological Sciences (KCBP)

WORK EXPERIENCE

Maaind, Neuroscience-AI startup, *R&D intern* June. 2024 – Aug. 2024

Decoded emotion from **human EEG** using unsupervised & supervised learning

RESEARCH EXPERIENCE

University of Cambridge, *Research assistant/associate* Nov. 2025 - Present

Interactions between two visual areas during visual detection in both head-fixed and freely moving mice

- Applied the Neuropixels Apollo system for large scale single-unit recordings

University of Cambridge, *PhD student* Oct. 2021 - Present

Optogenetic manipulations and electrophysiological recordings of both head-fixed and freely moving mice

- Performed simultaneous single-unit recordings in V1 and SC in freely moving mice
- Applied optogenetic inhibition to V1 and SC to assess their causal roles in visual detection
- Collaboratively developed a closed-loop system using DeepLabCut-live for controlling visual stimulus

Korea Institute of Science and Technology (KIST), *Graduate researcher* Dec. 2017 – Aug. 2020

Effect of social condition on amygdala under robot-induced threat

- Designed an innovative Robot-Induced Escape paradigm and discovered a new phenomenon
- Collaboratively developed the wireless neural recording system
- Validated the wireless recording system using rodent hippocampus and human EEG

SKILLS

Computational Skills

- **MATLAB** *(7+ years of experience)* and **Python** *(3+ years of experience)*
  Data analysis, for example, analysis of behavior, LFP and multi-channel spiking activity data, spike sorting using Kilosort and Phy, FFT/power spectrogram analysis.
- **Machine Learning and Deep Learning** PyTorch, CNN, DeepLabCut
- **Software** OpenSCAD, Photoshop, Illustrator, Word, Excel, PowerPoint

Biological Skills

Stereotaxic rodent surgeries; Viral injections; Design and 3D printing of custom devices, including optrodes for combined optogenetics and electrophysiology, and eye-tracking systems; Optogenetic stimulation; Perfusion, Behavior experiments (Fear conditioning, Tube test, Robot-Induced Escape Experiment, and Visual detection)

REFEREES

Dr Jasper Poort, Associate Professor Department of Physiology, Development and Neuroscience, University of Cambridge

Dr Riccardo Beltramo, Asisstant Professor Department of Physiology, Development and Neuroscience, University of Cambridge

Dr JeeHyun Kwag, Professor Department of Brain and Cognitive Sciences, Seoul National University

Dr Jee Hyun Choi, Professor Department of Bio-Medical Science and Technology, University of Science and Technology